HDOC DAY-45

Title: Date Formatting Transformation and Substring Generation

1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Change the date format from dd/mm/yyyy to dd-mmm-yy. E.g., Input: 12/10/2026, Output: 12-Oct-26.
2. Print all sub-strings of a string. E.g., Input: abc, Output: a,ab,abc,b,bc,c.
3. Exit.

2. Theoretical Background & Algorithm Logic

- **Date Format Conversion:**

The function parses integer values for day, month, and year from the input string using sscanf(). It uses a lookup array of month name abbreviations (Jan, Feb, ..., Dec) and extracts the last two digits of the year to format the output string as dd-mmm-yy.

- **Substring Generation:**

The function uses nested loops to iterate through all possible starting and ending indices of the input string, systematically printing every contiguous subsequence (substring) separated by commas.

3. Test Case Table

Test Case	Selected Option	Input Data	Expected Output	Actual Output	Status
1	Date Format	"12/10/2026"	"12-Oct-26"	"12-Oct-26"	Pass
2	Substrings	"abc"	"a,ab,abc,b,bc,c"	"a,ab,abc,b,bc,c"	Pass
3	Substrings	"xyz"	"x,xy,xyz,y,yz,z"	"x,xy,xyz,y,yz,z"	Pass
4	Exit	Choice 3	Program terminates cleanly	Program terminates cleanly	Pass

4. Source Code Listing

```
#include <stdio.h>
#include <string.h>
```

```
#include <stdlib.h>

void read_string(const char *prompt, char str[], int max_size) {
   printf("%s", prompt);
   scanf(" %[^\n]", str);
}

void convert_date_format(const char date_str[], char formatted_date[]) {
   int dd, mm, yyyy;
   if (sscanf(date_str, "%d/%d/%d", &dd, &mm, &yyyy) != 3) {
      strcpy(formatted_date, "Invalid Date Format");
      return;
   }

   const char *months[] = {"", "Jan", "Feb", "Mar", "Apr", "May", "Jun",
                           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
   if (mm < 1 || mm > 12) {
      strcpy(formatted_date, "Invalid Month");
      return;
   }

   int yy = yyyy % 100;
   sprintf(formatted_date, "%02d-%s-%02d", dd, months[mm], yy);
}

void print_substrings(const char str[]) {
   int len = (int)strlen(str);
   printf("Substrings: ");
   int first = 1;

   for (int i = 0; i < len; i++) {
      for (int j = i; j < len; j++) {
         if (!first) printf(",");
         for (int k = i; k <= j; k++) {
            printf("%c", str[k]);
         }
         first = 0;
      }
   }
   printf("\n");
```

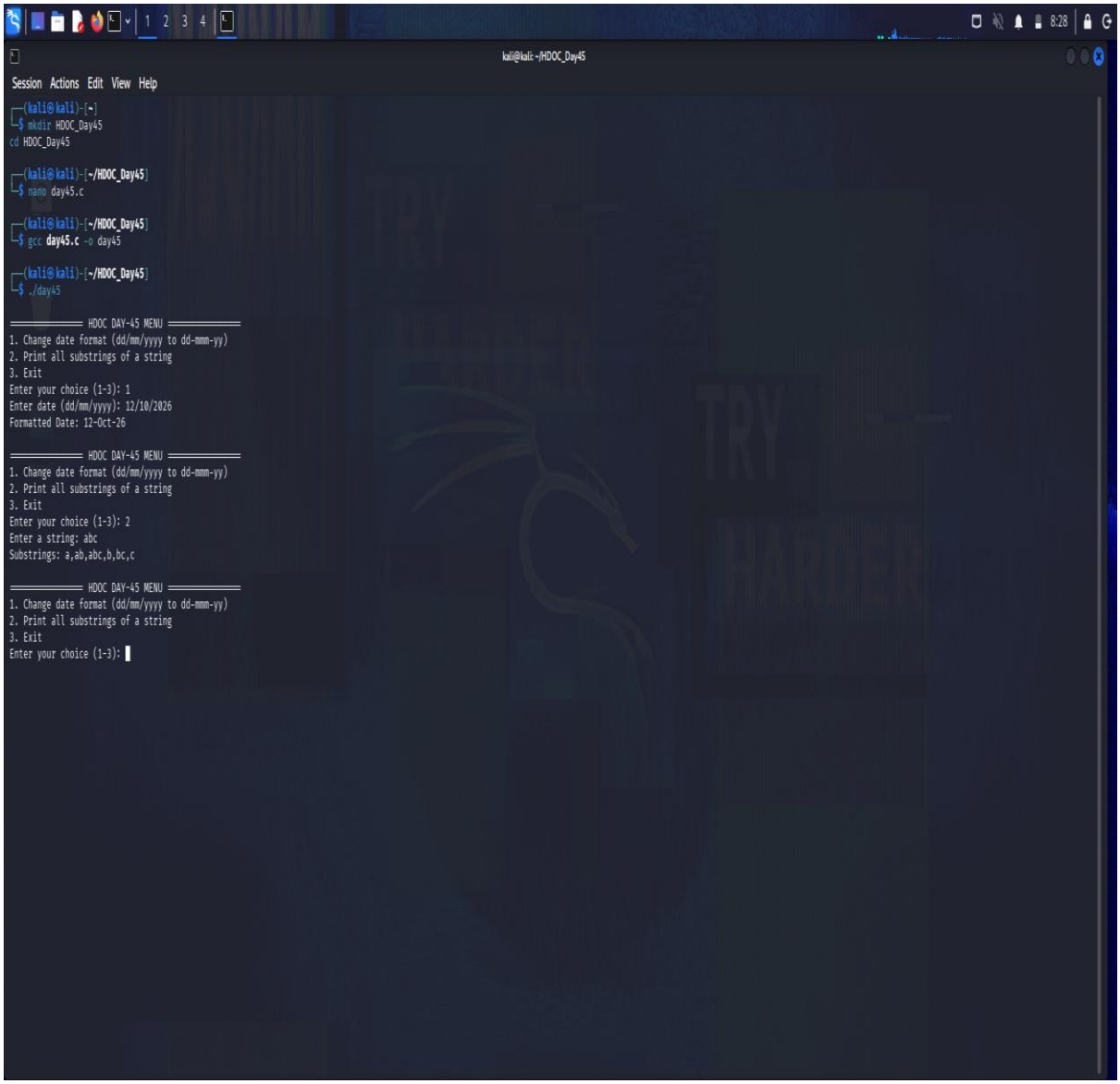
```
}

int main(void) {
   char date_in[20], date_out[50], str[100];
   int choice;

   do {
      printf("\n1. Date Format Conversion\n2. Print Substrings\n3.
Exit\nChoice: ");
      if (scanf("%d", &choice) != 1) break;

      switch (choice) {
         case 1:
            read_string("Enter date (dd/mm/yyyy): ", date_in, sizeof(date_in));
            convert_date_format(date_in, date_out);
            printf("Formatted Date: %s\n", date_out);
            break;
         case 2:
            read_string("Enter a string: ", str, sizeof(str));
            print_substrings(str);
            break;
         case 3: printf("Exiting...\n"); break;
         default: printf("Invalid choice!\n"); break;
      }
   } while (choice != 3);

   return 0;
}
```


kali@kali: ~/HDOC_Day45
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ mkdir HDOC_Day45
cd HDOC_Day45
┌──(kali㉿kali)-[~/HDOC_Day45]
└─$ nano day45.c
┌──(kali㉿kali)-[~/HDOC_Day45]
└─$ gcc day45.c -o day45
┌──(kali㉿kali)-[~/HDOC_Day45]
└─$ ./day45
═══════ HDOC DAY-45 MENU ═══════
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
Enter your choice (1-3): 1
Enter date (dd/mm/yyyy): 12/10/2026
Formatted Date: 12-Oct-26
═══════ HDOC DAY-45 MENU ═══════
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
Enter your choice (1-3): 2
Enter a string: abc
Substrings: a,ab,abc,b,bc,c
═══════ HDOC DAY-45 MENU ═══════
1. Change date format (dd/mm/yyyy to dd-mmm-yy)
2. Print all substrings of a string
3. Exit
Enter your choice (1-3):